

Generalized Policy Iteration

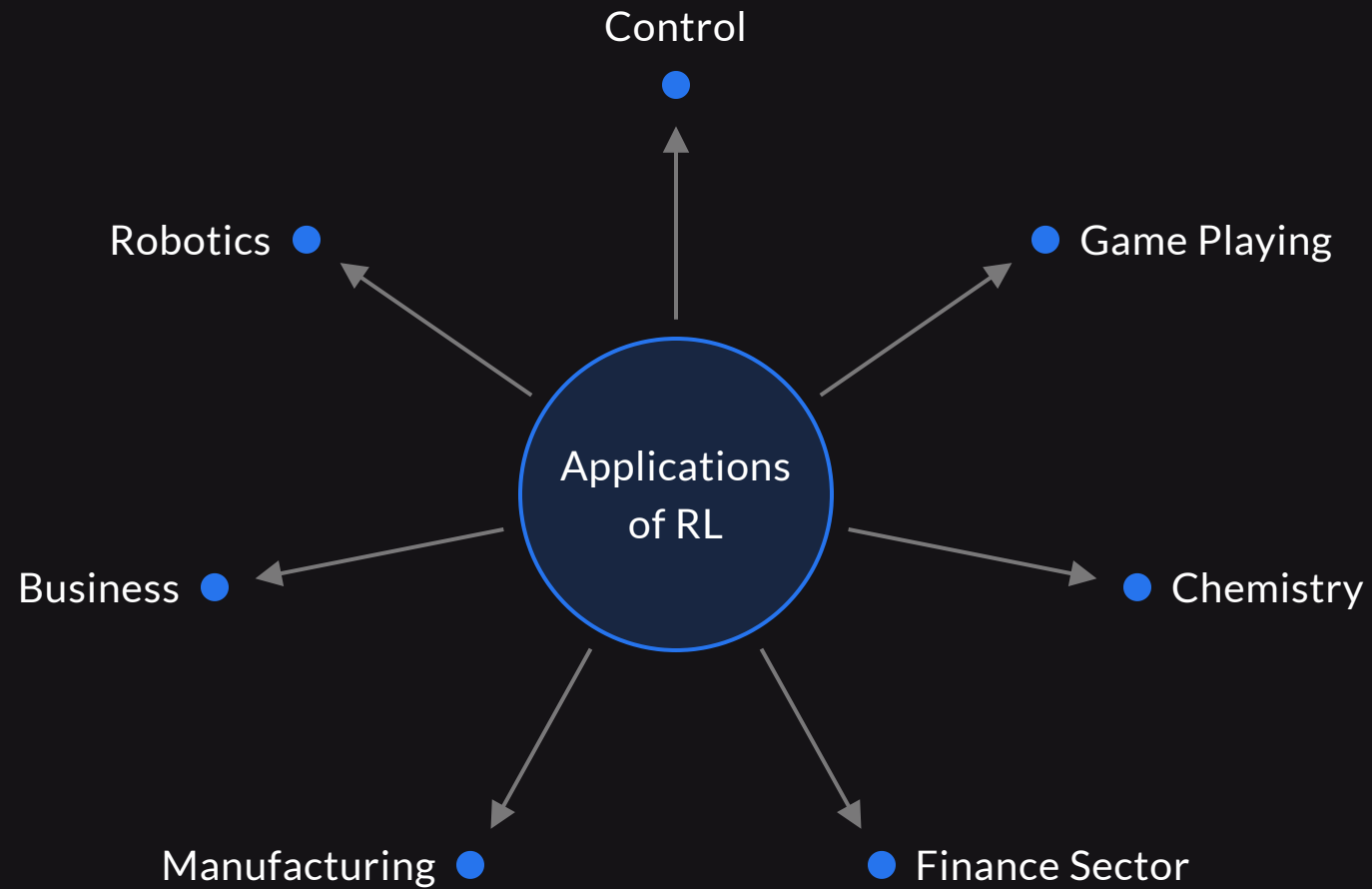
Instructor

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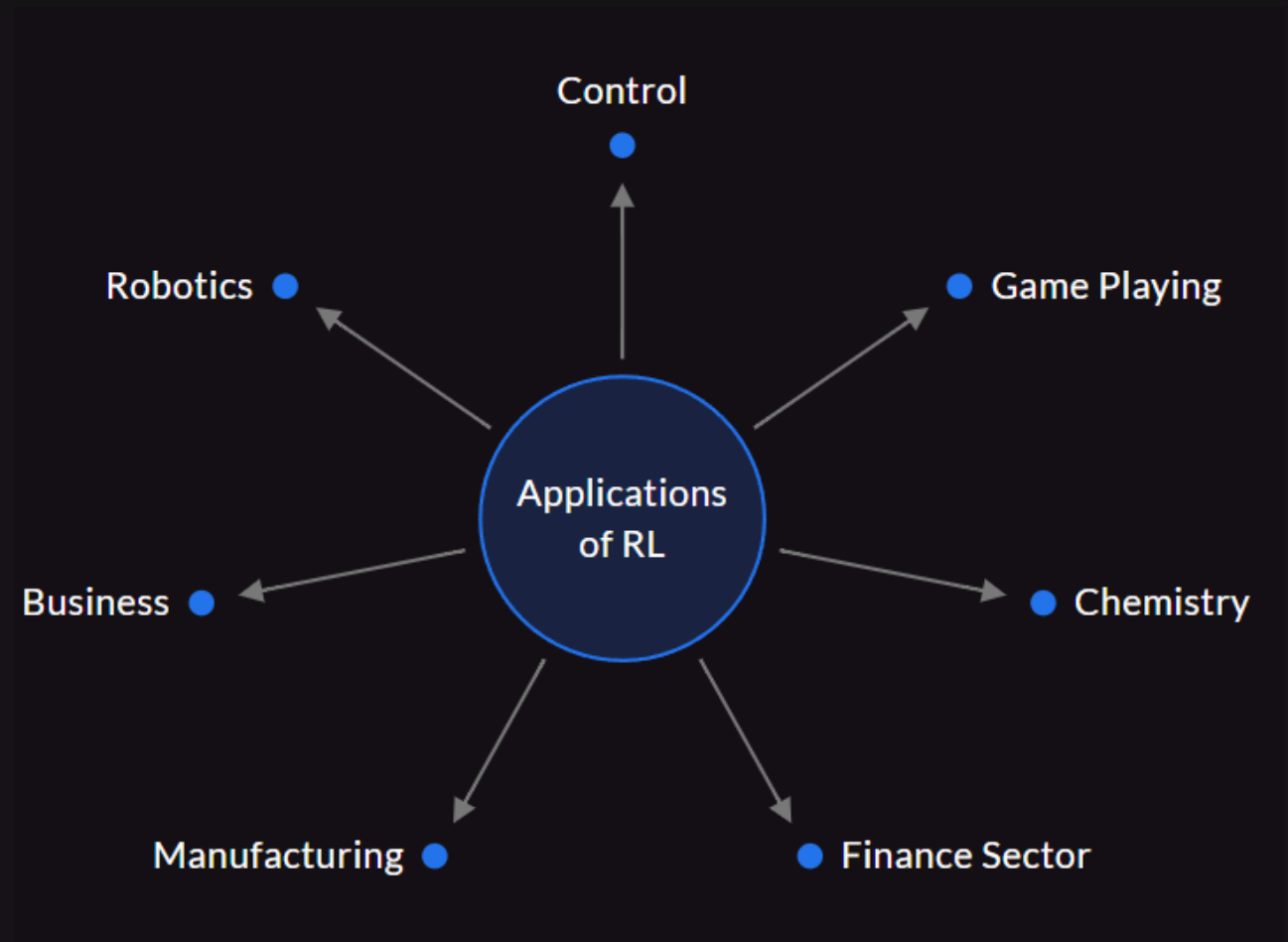
Model-Free Control



Model-Free Control

For most of these real-world problems, we have two challenges:

- Full MDP is unknown (i.e., model is unknown)
- Even if it is known, the state-space is significantly large
 - Cannot explore all possible transitions/rewards
 - Uses sampling



How do we solve such problems?

Model-free control

On and Off-Policy Learning

- Behavior policy:
 - Refine a single policy π that also generates control actions within the environment
 - Learn on the job
- Target policy:
 - Data from the behavior policy μ trains a separate target policy π for an optimization objective
 - Look over someone's shoulder

On-Policy Learning

Behavior policy = Target Policy

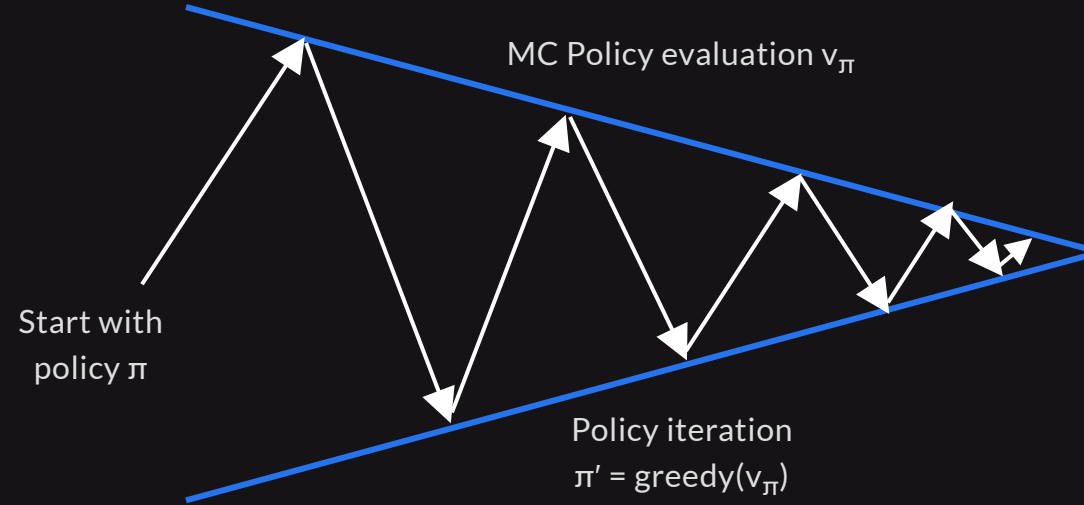
On and Off-Policy Learning

- Behavior policy:
 - Refine a single policy π that also generates control actions within the environment
 - Learn on the job
- Target policy:
 - Data from the behavior policy μ trains a separate target policy π for an optimization objective
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Off-Policy Learning

Behavior policy $\neq$ Target Policy

Generalized Policy Iteration with MC Evaluation



- Policy evaluation:

Can we use MC to evaluate policy? $v=v_{\pi}$

- Policy improvement:

Can we use greedy approach for policy improvement?

Greedy Policy Improvement in Model-Free Setting

- Problem with state value function:

Requires full knowledge of MDPs

$$\pi'(s) = \arg \max_{a \in \mathcal{A}} (R_s^a + \gamma \mathcal{P}_{ss'}^a v(s'))$$

Greedy Policy Improvement in Model-Free Setting

- Problem with state value function:

Requires full knowledge of MDPs

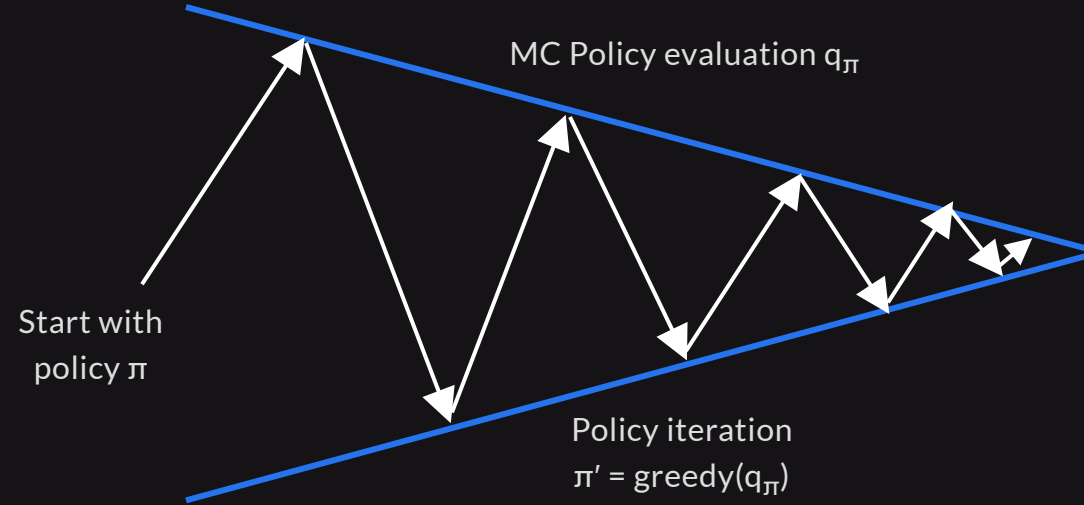
$$\pi'(s) = \arg \max_{a \in \mathcal{A}} (R_s^a + \gamma \mathcal{P}_{ss'}^a v(s'))$$

- Use action value function:

Use of $q(s, a)$ makes it model-free

$$\pi'(s) = \arg \max_{a \in \mathcal{A}} q(s, a)$$

Generalized Policy Iteration with MC Evaluation



- **Policy evaluation:**
Can we use MC to evaluate policy? $q = q_{\pi}$
- **Policy improvement:**
Can we use greedy approach for policy improvement?

Issues with Greedy Improvement



Behind door A is a successful career
in GenAI – behind the others is
tabulating Excel

Suppose there are three doors in front of you

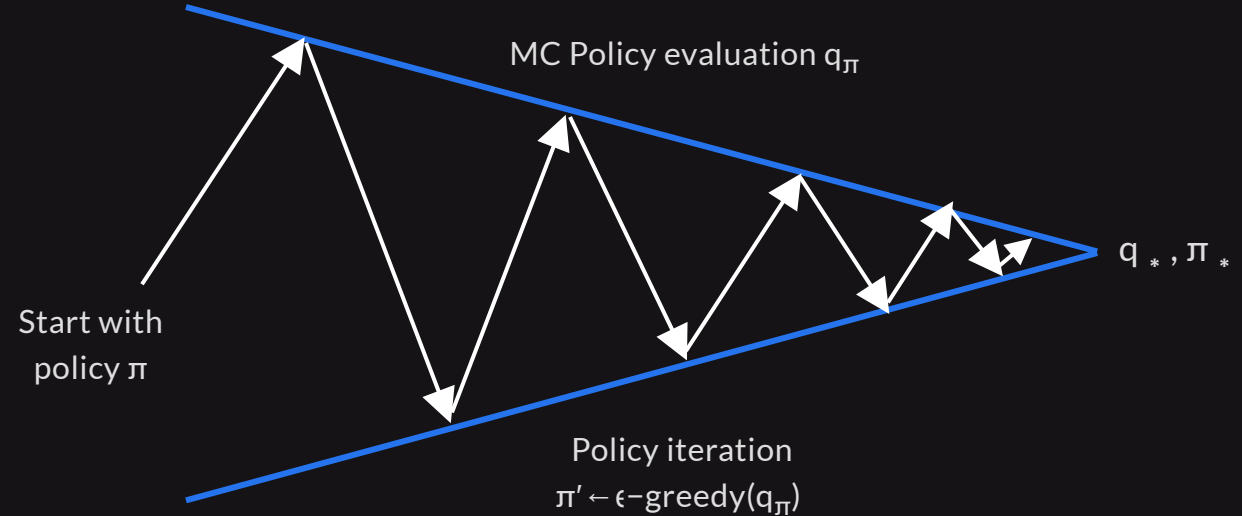
- You open door B and get a reward of -2
- You open door A and get a reward of +3
- You open door C and get a reward of -1
- You open door A and get a reward of +1
- You open door A and get a reward of +2

ϵ -Greedy Exploration

- Basic idea: Randomize action with a small probability ϵ and act greedily with a probability $1-\epsilon$
- **Explore** with probability ϵ and **Exploit** with probability $1-\epsilon$
- All m actions are tried with non-zero probability

$$\pi(a|s) = \begin{cases} \frac{\epsilon}{m} + 1 - \epsilon, & \text{if } a = \arg \max_{a' \in \mathcal{A}} q(s, a') \\ \frac{\epsilon}{m}, & \text{otherwise} \end{cases}$$

Monte-Carlo Policy Iteration



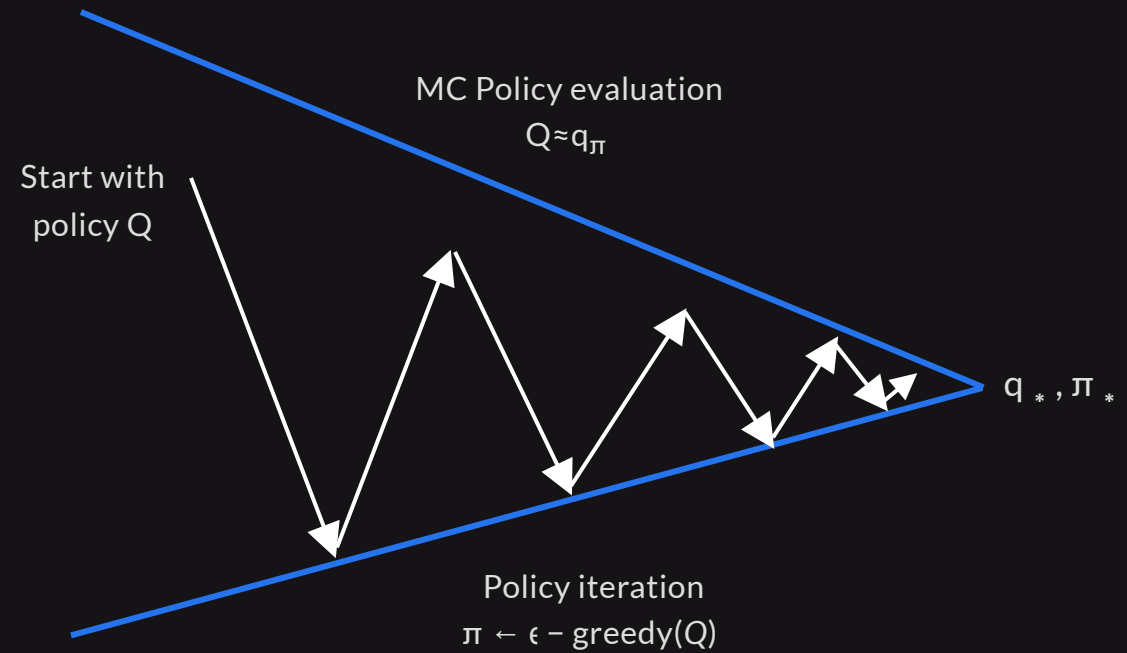
- Policy evaluation:

Can we use MC to evaluate policy? $q = q_{\pi}$

- Policy improvement:

Can we use greedy approach for policy improvement? $\epsilon\text{-greedy}$

Monte-Carlo Control



- **Policy evaluation:**
MC policy evaluation $Q \approx q_\pi$
- **Policy improvement:**
 ϵ -greedy policy improvement

Monte-Carlo Control

- For each episode k :
 - Sample using policy $\pi : \{S_1, A_1, R_2, \dots, S_T\} \sim \pi$
 - For each state-action pair (S_t, A_t) :

- Increment counter $N(S_t, A_t)$:

$$N(S_t, A_t) \leftarrow N(S_t, A_t) + 1$$

- Update $Q(S_t, A_t)$:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{1}{N(S_t, A_t)}(G_t - Q(S_t, A_t))$$

- Policy improvement:

$$\epsilon \leftarrow \frac{1}{k}$$
$$\pi \leftarrow \epsilon\text{-greedy}(Q)$$

Thank You
